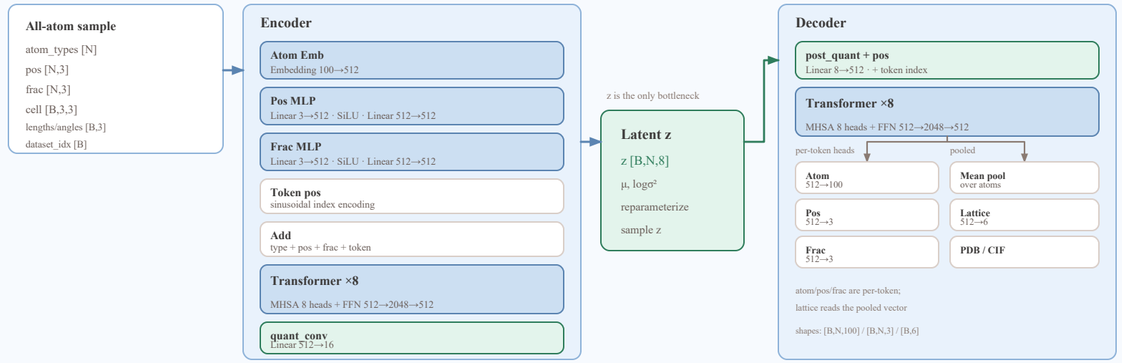
ADiT: All-atom Diffusion Transformer

Stage 1 all-atom VAE | Stage 2 flow-matching DiT-B | module-level view

forward · loss · frozen · VAE layer · DiT block

Stage 1 · VAE

8+8 Transformer layers · enc 25.8M + dec 25.3M · latent 8



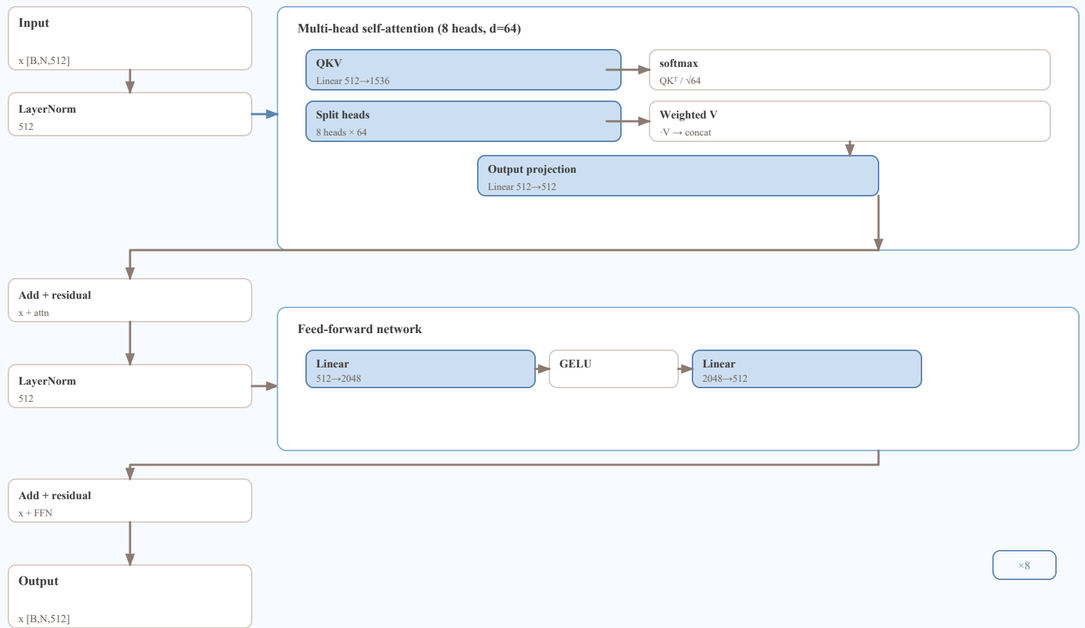
no skip connections; encoder and decoder are separate stacks

L_VAE = CE(atom types) + MSE(lengths/angles/frac/pos) + 1e-5·KL

denoise: 10% atom-type mask + 10% coordinate noise

VAE Transformer layer

shared by encoder and decoder · ×8 each · pre-norm residual

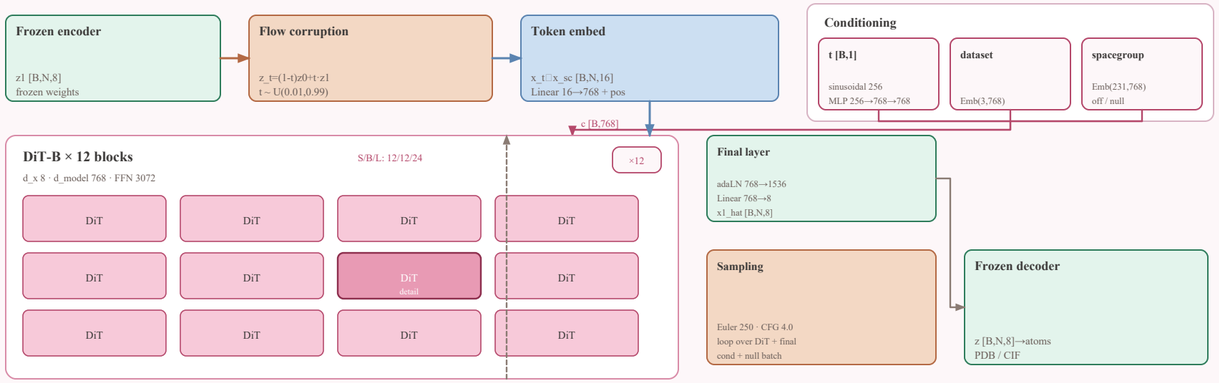


dropout = 0 · norm_first = True · residual per sublayer

decoder uses the same non-causal TransformerEncoder stack

Stage 2 · DiT-B

12 DiT blocks · d_model 768 · 12 heads · 129.7M

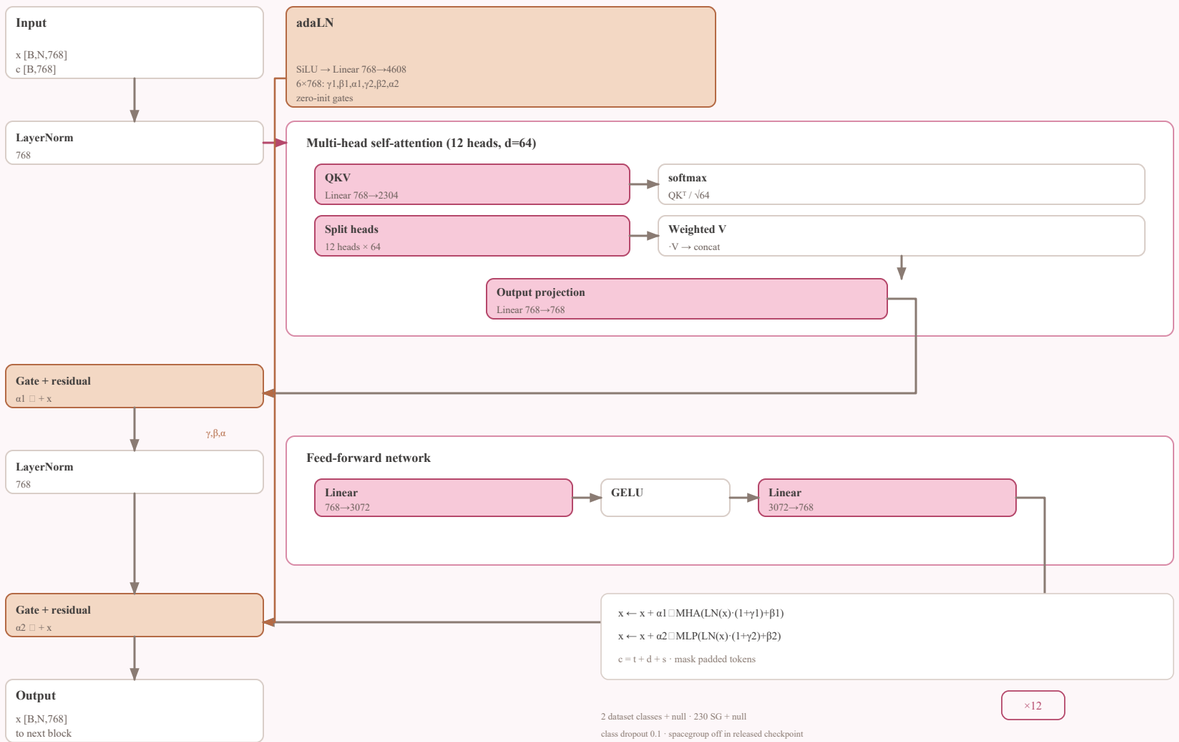


L_FM = mean[mask · ((x1 - x1_hat)/(1-min(t,0.9)))²]

mask = valid tokens; VAE frozen, only DiT updates

DiT block

one of 12 blocks · ≈10.63M/block · adaLN conditioning · ×12



2 dataset classes + null · 230 SG + null

class dropout 0.1 · spacegroup off in released checkpoint